# Direction fields and solution curves

We've been looking at different forms of first order differential equations, and we've been learning analytical methods for finding the equations of their solutions.

But it's important to remember that, in the real world, differential equations often won't behave so nicely. In other words, it's common to face differential equations that don't match one of our perfect formats (linear, separable, Bernoulli, homogeneous, exact, etc.). It's also normal to see equations that have a solution that we can't use an analytical method to find, or even a differential equation that has no solution at all.

But even when there isn't a solution, or there *is* a solution but we can't calculate it, we can still often draw conclusions about the behavior of the differential equation.

So how will we find the solution to a differential equation that isn't one of these special types? Well, instead of taking an analytical approach, we can take a geometric approach.

## A geometric approach

A **direction field** is a graph made of many small arrows, each of which approximates the slope of the curve in the area where the arrow is plotted.

Up to now, we've been looking at differential equations containing $y'$, or $dy/dx$, and solving them analytically to find a solution $y$, or $y(x)$. A direction

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field is just a geometric way of displaying the same information in the differential equation, and sketching a solution curve, also called an integral curve, through the direction field is just a geometric way of displaying the same information in the solution equation.

In other words, writing the equation of the differential equation $y'(x)$ is the same as sketching its direction field, and finding the solution $y(x)$ is the same as sketching the solution curve through the direction field. They're just two different ways of displaying the same set of information.

So a direction field is a sketch of a differential equation, and an integral curve plotted through the direction field is the sketch of a solution to that differential equation.

## Sketching direction fields

Our first approach to creating a direction field by hand will be to choose a set of equally spaced coordinate points, and then calculate the derivative at each point.

Once we find the value of the derivative at each coordinate point, we'll sketch a tiny arrow there, whose slope is equal to the value of the derivative.

Repeating this process across many coordinate points will create the direction field. If the direction field doesn't give us enough detail to get a good idea of how the derivative is behaving, then we can go back through and calculate the derivative at additional coordinate points.

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And once we have a good direction field sketched out, we can use it to sketch a solution curve through any coordinate point. Each of the little arrows in the direction field represents the tangent to the solution curve at that point. So sketching the solution curve means following the direction indicated by the arrows, keeping the solution curve tangent to each of the arrows.

Of course, we're only doing this to see how a direction field is actually built, but once we understand the concept, going forward we'll use a calculator or computer to quickly get a picture of the direction field when we need it.

Let's look at an example so that we can see how to build the direction field for a differential equation.

**Example**

Sketch a direction field for the differential equation, then sketch a solution curve that passes through $(1,1)$.

$$y' = y - x$$

For a differential equation in this form, we'll sketch the direction field by using coordinate pairs $(x, y)$ to find corresponding values for $y'$. For example, if we take the coordinate pair $(x, y) = (-2, -2)$, we can plug it into the differential equation to get

$$y' = y - x$$

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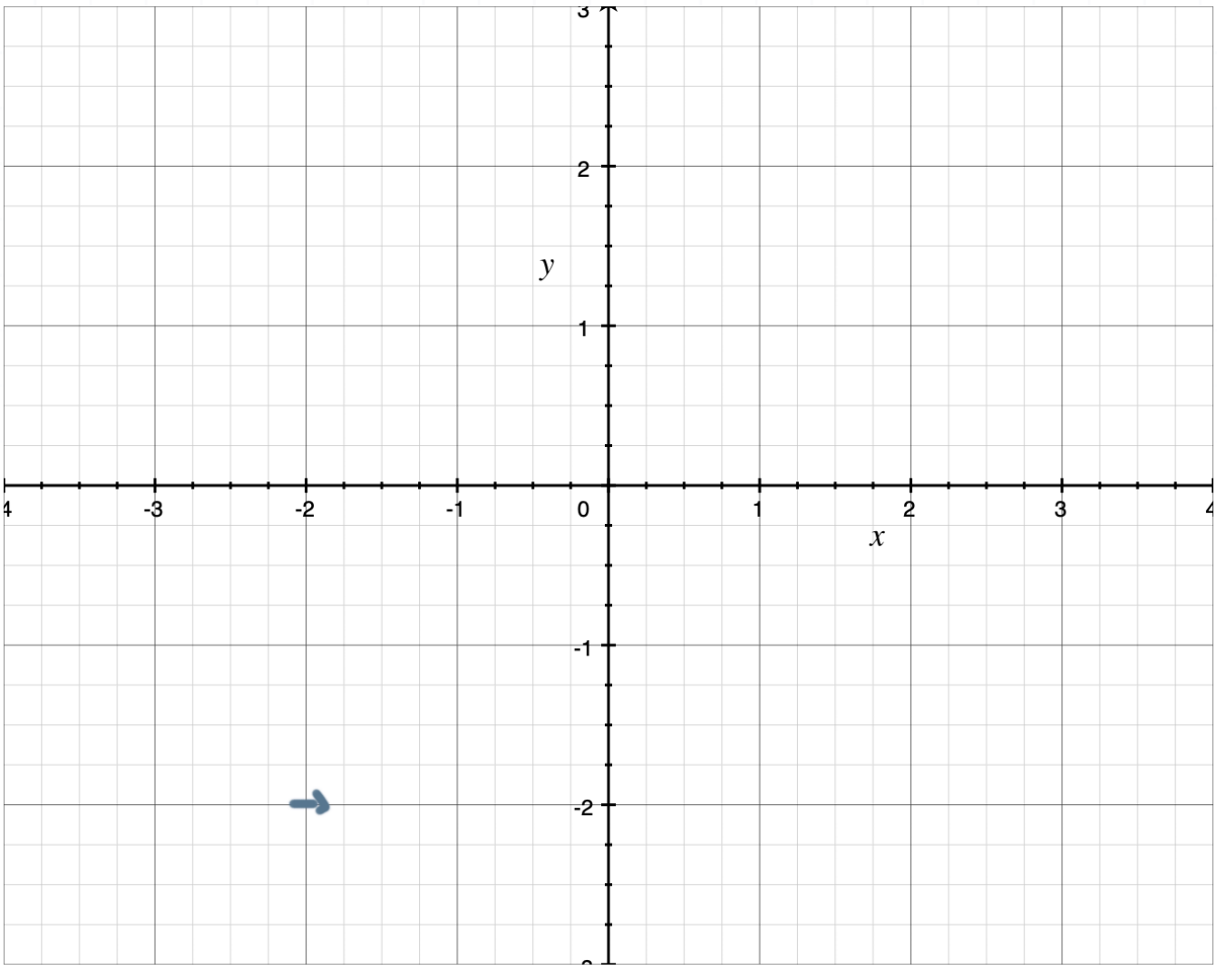


$$y' = -2 - (-2)$$

$$y' = -2 + 2$$

$$y' = 0$$

Since $y'$ is the derivative (the slope of the tangent line), we can say that the slope at $(-2, -2)$ is $0$. To sketch this information into the direction field, we'll draw a very small arrow with slope $0$ (so the arrow will be horizontal), at the point $(-2, -2)$.




The trick with these kinds of problems is keeping all this information organized. We'll start by picking values for $x$, like $x = \{-2, -1, 0, 1, 2\}$. Then we'll choose the same values for $y$, and we'll pair each value of $y$ with each of the values we chose for $x$. Since we have five values for $x$,

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$x = \{-2, -1, 0, 1, 2\}$, and the same five values for $y$, $y = \{-2, -1, 0, 1, 2\}$, we'll get $5 \times 5 = 25$ coordinate pairs. We'll plug each of these $25$ points into the differential equation to find an associated value for $y'$.

We can record all of these values in a single table, placing $x$-values across the top row, $y$-values down the first column, and the resulting $y'$ values in the body of the table.

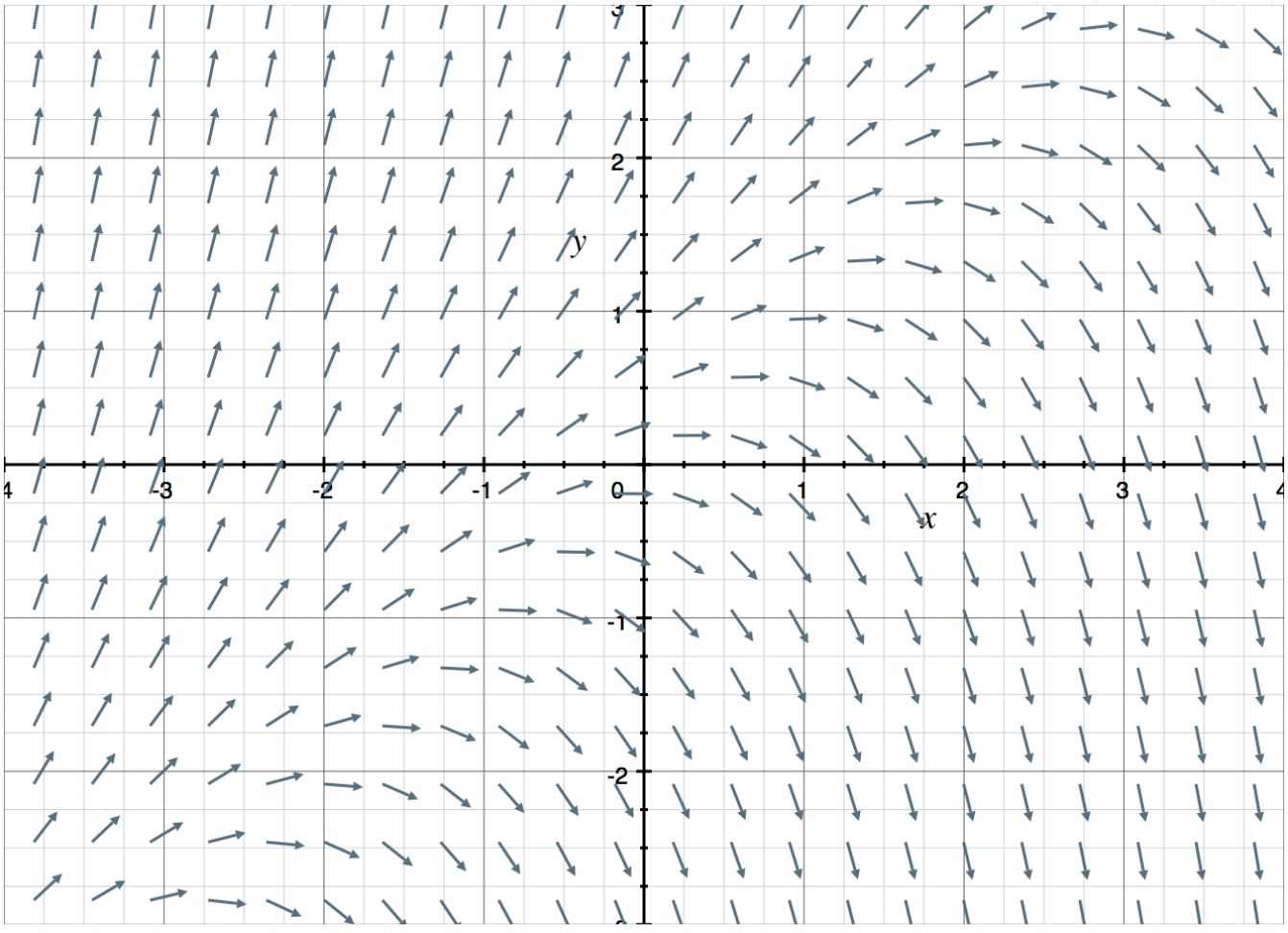
| | | x | | | | |
|---|---|---|---|---|---|---|
| | | **-2** | **-1** | **0** | **1** | **2** |
| **y** | **-2** | 0 | -1 | -2 | -3 | -4 |
| | **-1** | 1 | 0 | -1 | -2 | -3 |
| | **0** | 2 | 1 | 0 | -1 | -2 |
| | **1** | 3 | 2 | 1 | 0 | -1 |
| | **2** | 4 | 3 | 2 | 1 | 0 |

To sketch this information into the direction field, we navigate to the coordinate point $(x, y)$, and then sketch a tiny arrow that has a slope equal to the $y'$-value. Remember, if $y' = 0$, the small arrow will be horizontal. If $y'$ is positive the arrow will tilt up to the right, and if $y'$ is negative the arrow will tilt down to the right. The larger the value of $y'$, either positive or negative, the steeper (closer to vertical) the arrow will be. The smaller the value of $y'$, either positive or negative, the shallower (closer to horizontal) the arrow will be.

The more $y'$ values we calculate, the more accurate our direction field will be. Here's a detailed direction field for $y' = y - x$:

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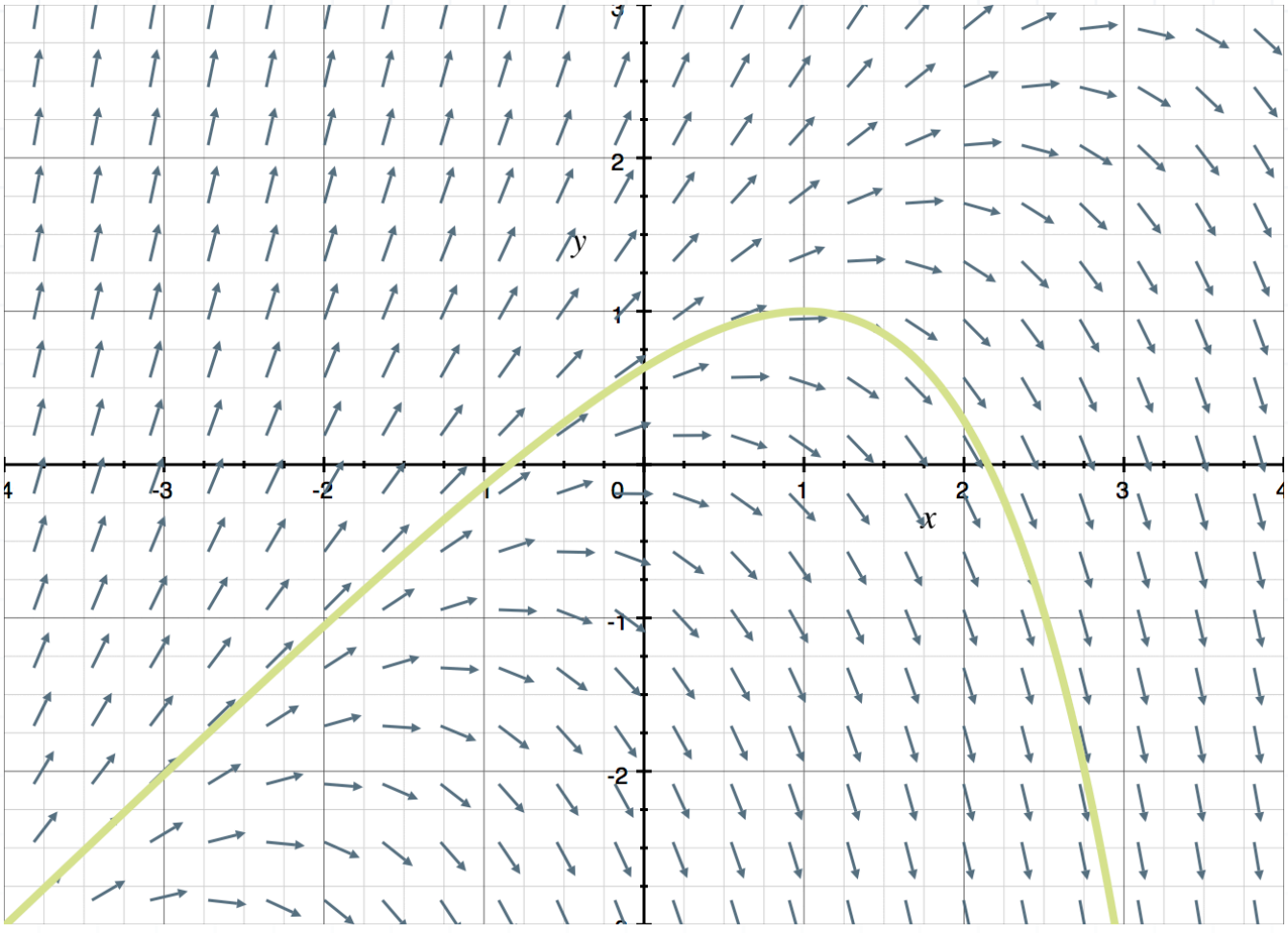





To address the second part of the question, we'll sketch the solution curve that passes through $(1,1)$. To do that, we'll plot a point at $(1,1)$. From there, we'll follow the direction arrows near that point as closely as we can, trying to ensure that the curve we sketch is tangent to all of the arrows we pass by.

If we keep sketching in the direction that the nearby arrows indicate, we get a rough picture of the solution curve through $(1,1)$.

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A solution curve that we sketch by hand will never be exact. Instead, it's an approximation we create as we follow the small arrows in our direction field that indicate the slope of the curve near each arrow.

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But there's another way to sketch a direction field, which is actually a little more efficient to do by hand, which is to choose a certain value of $y'$ (choose a certain slope) and then find the equation of the curve that maintains that slope.

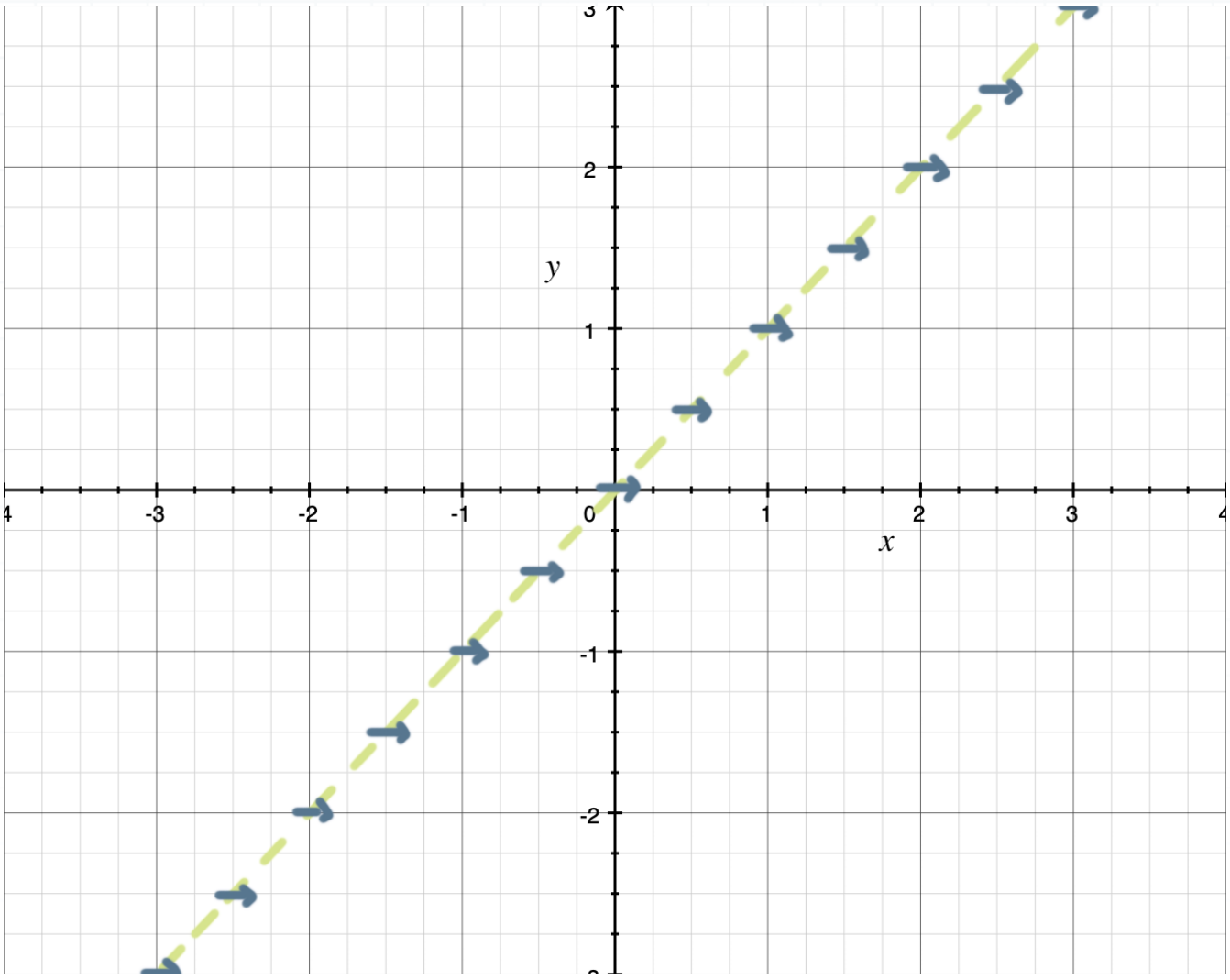
For example, looking at the differential equation $y' = y - x$ from the last example, if we choose $y' = 0$, then the differential equation can be rewritten as

$$0 = y - x$$

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$$y = x$$

This result tells us that the derivative is always $0$ along the line $y = x$.



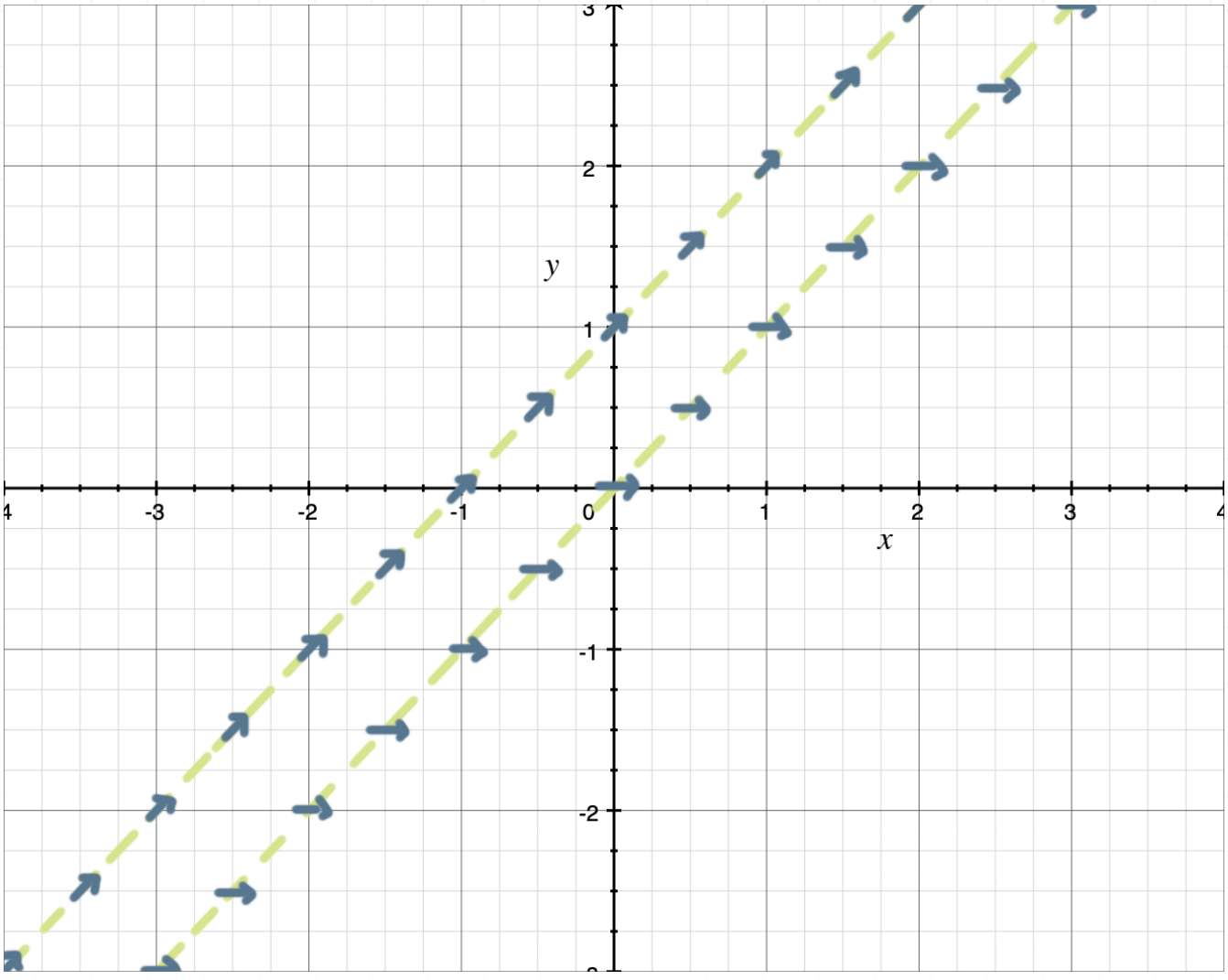

If we choose $y' = 1$, then the differential equation can be rewritten as

$$1 = y - x$$

$$y = x + 1$$

This result tells us that the derivative is always $1$ along the line $y = x + 1$.

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And in general, if we choose any $y' = c$, then the differential equation can be rewritten as

$$c = y - x$$

$$y = x + c$$

In this form, we can quickly pick any slope $c$, then sketch that slope along the line $y = x + c$. The family of curves we get by choosing different values for $c$ in $y = x + c$ are the **isoclines** of the equation, so we can say that the derivative will be equal to the constant $c$ along any isocline of the equation.

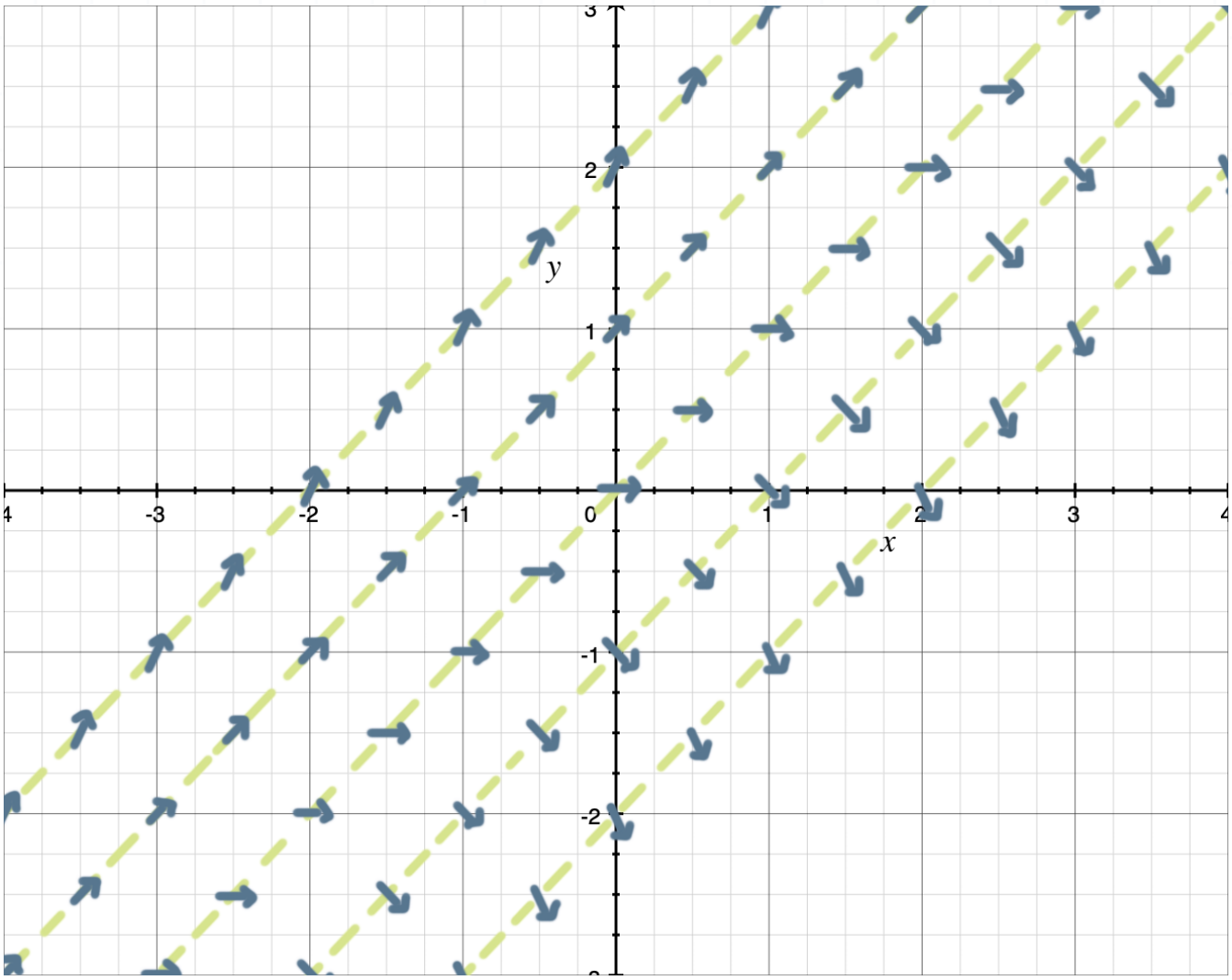
Which means that, for this equation, we can

- sketch tangents with slope $-2$ along the isocline $y = x - 2$

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- sketch tangents with slope $-1$ along the isocline $y = x - 1$
- sketch tangents with slope $0$ along the isocline $y = x$
- sketch tangents with slope $1$ along the isocline$y = x + 1$
- sketch tangents with slope $2$ along the isocline $y = x + 2$
- ...

Using this method, we can more quickly put together the direction field.




And we can sketch solution curves in the same way we did before, which is by following the slopes in the direction field.

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Of course, if it's possible to find the equation of the solution curve by solving the differential equation (in this case we can solve the equation as a linear differential equation),

$$y' = y - x$$

$$y' - y = -x$$

$$I = e^{-x}$$

$$y'e^{-x} - ye^{-x} = -xe^{-x}$$

$$\frac{d}{dx}(ye^{-x}) = -xe^{-x}$$

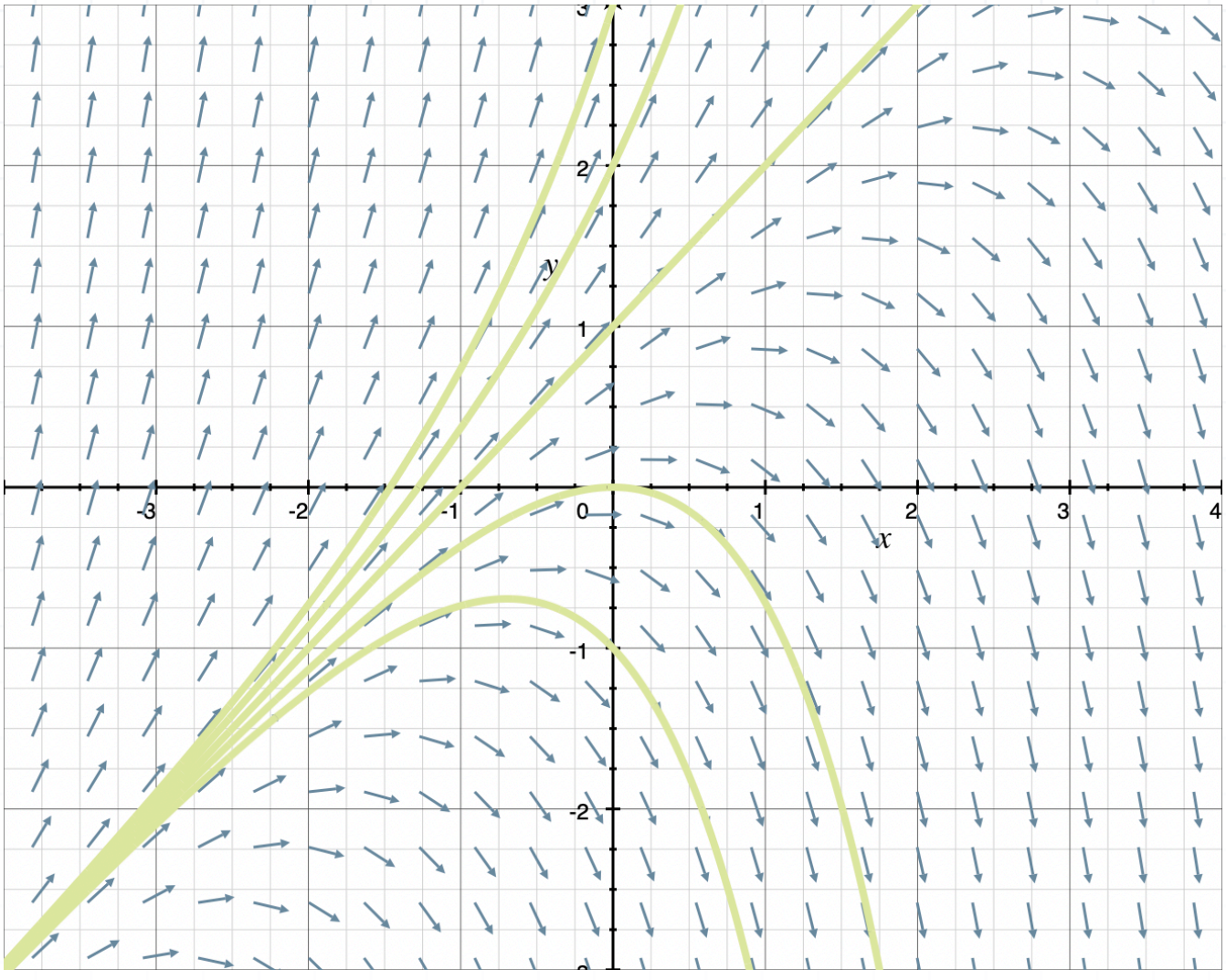
$$\int \frac{d}{dx}(ye^{-x})\,dx = \int -xe^{-x}\,dx$$

$$ye^{-x} = xe^{-x} + e^{-x} + C$$

$$y = x + 1 + Ce^{x}$$

then we can choose different values for the constant of integration $C$ to sketch different solution curves. Below are the solution curves for $C = \{-2, -1, 0, 1, 2\}$.

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